

PAKISTAN CYBERSECURITY ALLIANCE (PCA)

Managed by iSeeWaves (Private) Limited

THREAT HORIZONS PAKISTAN

Event Summary Whitepaper — Episode 1

A Cross-Sector Dialogue on Cybersecurity, AI & Governance



Hosted at: National Incubation Center for Aerospace Technologies - NICAT, Rawalpindi

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Background and Context

The Pakistan Cybersecurity Alliance (PCA) started with a simple idea: Pakistan's cybersecurity community should not have to work in silos. Professionals, academics, students, and policymakers all care about the same problem; a resilient digital Pakistan, but rarely sit in the same room. PCA exists to change that.

PCA is a free, open, industry-led community: no paywalls, no gatekeeping, just monthly expert sessions, sector roundtables, and training pilots designed to build real technical capability while connecting local talent to what is happening globally in the field.

This particular event brought that mission to life. Hosted in partnership with iSeeWaves (Private) Limited and the National Incubation Center for Aerospace Technologies - NICAT in Rawalpindi, it pulled together voices from industry, academia, and policy to talk honestly about the cyber threats facing Pakistan's tech and financial sectors not in the abstract, but with specifics: real gaps, real risks, real cases.



Figure 1: Threat Horizons Pakistan — 9th July 2026, NICAT Rawalpindi

Vision

A Pakistan that is not just defending against cyber threats, but shaping how the world thinks about them. A place where skilled talent, sound policy, and strong digital defenses work together to protect the country's economy, infrastructure, and people.

Mission

PCA works to close the distance between the people who study cybersecurity, the people who practice it, and the people who write policy around it. At its core, PCA is betting that the fastest way to make Pakistan's digital ecosystem safer is to stop treating cybersecurity as any one sector's job and start treating it as everyone's.

Keynote Speaker 1: Mr. Abdullah Nasir

“The Missing Umbrella”

Abdullah Nasir, CEO of iSeeWaves (Private) Limited and President of the Pakistan Cybersecurity Alliance, opened the event with a talk that set the tone for everything that followed. His central message was simple and blunt: *we are all threads now, woven into the same fabric, whether we planned for it or not.*

He backed up his argument with figures that made the room sit up, and framed the last three decades as a steady march toward total interdependence across sectors that once stood apart.



Mr. Abdullah Nasir
CEO, iSeeWaves (Private) Limited
President, PCA

The Numbers

Metric	Figure	Source
Year-over-year escalation in total cyber incidents (Pakistan)	35%	National Cyber Crime Investigation Agency
Increase in third-party supply-chain breaches	Doubled	Verizon 2025 DBIR
Attacks on Pakistani infrastructure, Q1 2026	98+	Senate Standing Committee
Projected global cost of cybercrime, 2026	US\$11.88 trillion	Proxymark Global Cybercrime Report

Table 1: Key statistics cited during the opening keynote.

A Decade-by-Decade Shift

Period	State of Connectivity
1990s	Systems; factories, hospitals, banks, power grids stood largely on their own.
2010s	Sectors began connecting into small clusters.
Today	Everything links into one dense web: one interdependent economy in which a car, a hospital, a bank, and a power station are all nodes on the same network.

Table 2: Evolution of cross-sector interdependence, as presented by Mr. Nasir.

The Missing Umbrella

That interdependence is the problem but it is also the shape of the solution. If everyone is exposed together, everyone has to defend together. Nasir called this “the missing umbrella”: the shared protection that Pakistan’s cybersecurity ecosystem still lacks.

“The umbrella starts with us. One nation, one defense. No sector stands alone.”

— **Abdullah Nasir, President, Pakistan Cybersecurity Alliance**

His proposed fix rested on four points:

- **One room, one defense:** no sector faces this alone
- **Connect every sector’s hands:** shared intelligence across industries
- **Treat threats and solutions as shared,** not siloed
- **Use PCA as the starting point:** the room they were building together, starting that day

Keynote Speaker 2: Ms. Quratulain Chaudhary

Securing the Financial Ecosystem Through Industry, Tech & Academic Alliances”



Ms. Quratulain Chaudhary

VP, Women Financial Solutions & Advocacy

Mobilink Microfinance Bank (MMBL)

Quratulain Chaudhary, Vice President of Women Financial Solutions and Advocacy at Mobilink Microfinance Bank (MMBL), opened with a line that framed her entire talk: *trust is Pakistan's newest financial currency*. She grounded that claim in MMBL's own growth, then pivoted to a sober warning: the pace of financial inclusion is outrunning the guardrails meant to protect it.

Three Risks Outrunning Guardrails

Growth, she warned, is outrunning guardrails. She laid out three specific risks:

1. **Widening attack surface:** every new digital account, wallet, and app is a new front door, and Pakistan is opening millions of them a year.
2. **Social engineering as the fastest-growing fraud vector:** it targets people rather than systems, hitting the least digitally literate customers hardest.
3. **First-time users as first-time targets:** newly banked customers, disproportionately women, combine the least fraud pattern-recognition with the highest institutional trust.

MMBL at Scale

Indicator	Figure
Active female customers	19 million
Active women's financing portfolio	PKR 50.3 billion
Year-over-year portfolio growth	41%
Share of female customers overall	Grew from 15% to 27%

Table 3: MMBL scale metrics; internal records and WEF Future of Inclusion Lighthouses 2026.

A Three-Pillar Framework: Industry, Technology, Academia

Industry

MMBL is one of 22 banks committed to the State Bank of Pakistan's WE Finance Code and is the only Pakistani institution in the WEF Future of Inclusion Lighthouses 2026 cohort. Her principle: an alliance is only as strong as the data-sharing and reporting standards its members hold each other to.

Technology

She highlighted the BIBI App (co-developed with Shifa International Hospital) and the NIC Sialkot / WISE hub. Her principle: technology earns adoption through trust, and in digital finance, trust is a security outcome before it is a marketing one.

Academia

She pointed to the WIN Incubator (31+ graduates across two cohorts with universities and CARE International) and a research partnership with SDPI. Her principle: academia supplies not just graduates but the independent evidence that keeps industry honest.

Three Asks for the Alliance

1. A **shared fraud-signal registry** for industry-wide intelligence sharing across banks, microfinance banks, and NBFCs.
2. **Security modules built directly into literacy programs**, so every session doubles as a fraud-awareness session.

3. A **joint industry-academia research agenda**, giving universities real transaction-pattern questions and institutions evidence-based defenses in return.

“A financial ecosystem is only as inclusive as it is secure, and only as secure as the alliances behind it.”

— **Ms. Quratulain Chaudhary, VP MMBL**

Panel Talk: Bridging the Gap Between Academia, Security Policy, and Industry Practice

The panel was moderated by Ms. Rabia Khan, Founder of HerVoice and Assistant Registrar at the University of Greater Manchester Islamabad, whose background spans a master's in Computer Science (AI and ML), UN project management, and regular appearances on PTV Home and Suno Network on AI and Pakistan's digital future.

The conversation brought together three distinct vantage points, policy, academia, and industry, and pressed each panelist on where responsibility actually sits for closing Pakistan's cybersecurity capacity gap.



Panel Discussion

Moderator: Ms. Rabia Khan

Panelists: Khurrum Javed,
Abdullah Nasir, Abid Jameel

The Panelists

Panelist	Sector	Role
Mr. Khurrum Javed	Policy & Capacity	Director Capacity Building, National CERT Pakistan
Mr. Abid Jameel	Academia	Lecturer, Faculty of Computing, IIUI
Mr. Abdullah Nasir	Industry	CEO, iSeeWaves (Private) Limited

Table 4: Panelists and the sectors they represented.

A Shared Triangle, Not a Single Point of Failure

Mr. Khurrum Javed framed the discussion from the outset around collaboration rather than blame. He pointed to National CERT's ongoing work building practical-level training capacity for students, and argued that industry, academia, and government need to build this capacity together rather than in isolation. Faculty at

academic institutions, he said, need their own capacity-building initiatives, staying current with practical, industry-relevant knowledge so they can in turn train students able to perform and stand out in the field. University labs, FYPs, and student projects, he added, should be built around actual industry standards rather than generic coursework.

Mr. Abdullah picked up this thread and pushed it toward self-reliance. He called for a deliberate national effort, driven through FYPs and academic research, to build open-source, indigenous platforms suited to Pakistan's own local needs, so the country stops depending on foreign tooling and instead develops and accelerates its own products. He was direct about where responsibility sits: the current gap is not the fault of one sector alone. Government, industry, and academia all share it, and closing it requires treating the three as a single triangle rather than three separate problems.

Are Certifications Actually Working?

Asked directly whether the certifications students pursue are translating into real capability, Mr. Abdullah was blunt: certifications are largely a piece of paper. What matters, he argued, is the skill built while earning them, which is what industry actually demands. His proposed fix was practical rather than credential-based: students should use the process of certification to identify real problems in their own ecosystem, then apply what they've learned to build actual solutions, rather than treating the certificate itself as the end product.

Why the Classroom Doesn't Match the Real World

Pressed on whether the environment students train in resembles what they'll actually face, the panel was unanimous that it does not. The real world, once students enter it, looks nothing like the isolated academic environment they trained in. The proposed remedy echoed Mr. Khurram Javed's opening point: specialized labs and specialized curricula, built around National CERT's capacity-building mandate, are necessary to close that gap before graduation rather than after.

Mr. Abid Jameel located the disconnect more specifically at the instructor level. Faculty, he said, are already pushing students toward building commercializable products aimed at solving specific problems facing Pakistan. The barrier is not intent but support: Pakistan has researchers with 200 to 300 published papers, ranked among the best in the world, yet that research is largely going to waste. Government needs to actively back these researchers and developers, giving their work a path from published paper into a practical, productized form. Only once that happens does the harder task begin, getting those products into real-world use and commercializing

them, which he identified as equally critical to the country's growth.

What Each Sector Needs to Give Up

Taken together, the three panelists each pointed to a different piece of the same triangle:

- **Academia** needs labs, FYPs, and instructor knowledge that track industry standards, not isolated theoretical benchmarks.
- **Industry** needs to stop treating certifications as sufficient proof of skill and instead engage students on real, unsolved problems.
- **Government** needs to fund and support the path from published research to deployable, commercialized product, rather than leaving that research stranded.

Consensus and Next Steps

All three panelists, representing government, industry, and academia respectively, agreed that everyone needs to take at least the first step, and that this is not a problem any single sector can solve alone. Mr. Khurram closed by walking through several of National CERT's current initiatives, giving the room a clearer picture of where the country stands and what gaps remain. From there, the panel pointed to concrete opportunities already open to students, open-source contribution chief among them, as practical ways to start closing the gap directly rather than waiting on institutional fixes.

"Closing the gap isn't about academia, policy, and industry defending their own turf, it's about each one giving up a little control so the other two can move faster."

Keynote Speaker 3: Mr. Salman Dar

“AI or Be Left Behind”



Mr. Salman Dar

CEO, ITROOS (Private) Limited

Salman Dar, CEO of ITROOS (Private) Limited, delivered one of the day's sharpest warnings; aimed less at cybersecurity in the narrow sense and more at the survival of Pakistan's tech industry as a whole. His talk, titled “AI or Be Left Behind, The Risks Pakistan's Tech Industry Cannot Afford to Ignore,” opened by dismantling the industry's usual comfort blanket: the assumption that technology cycles every ten to fifteen years, so there is no need to worry this time either. This cycle, he argued, is different, not in kind but in scale and speed, and the impact is unprecedented compared to previous shifts.

He reframed the entire question the industry should be asking. The relevant question is not whether AI will have an impact, that much is settled, it is how quickly an organization or a country becomes irrelevant if it chooses to ignore it.

Pakistan's IT export industry has largely been built on a labor arbitrage model: cheaper developers, longer hours, service-based delivery. **AI attacks that model at its foundation**, the advantage Pakistan has competed on for years is precisely the advantage AI erodes first.

But Mr. Salman Dar was careful not to leave the room on a note of pure threat. He reframed the disruption as opportunity: Pakistan missed the hardware wave and the offshoring wave that built other economies, and by his account, AI is a fresh start, the first time in a generation the playing field has reset in Pakistan's favor. A five-person team in Islamabad can now compete with a Silicon Valley startup, but only if the industry stops selling hours and starts selling impact, moving from services to products.

He closed by returning to the urgency of his opening frame: AI will not wait for Pakistan to be ready. The choice facing the industry, as he put it, is stark and binary, remain the workforce of the past, or become the builders of the future.

“AI will not wait for Pakistan to be ready. The choice is stark: remain the workforce of the past, or become the builders of the future.”

— Mr. Salman Dar, CEO ITROOS

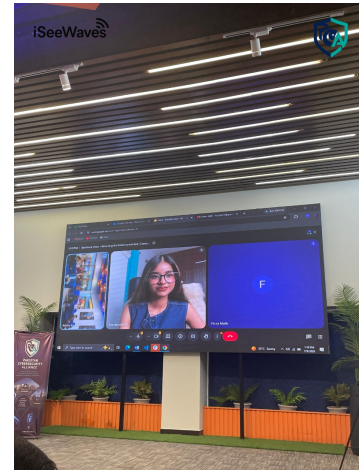
International Speaker 1: Ms. Bandana Kaur

“Securing the Interconnected: Cross-Sector Cyber and AI Risks in a Shared Economy”

Ms. Bandana Kaur, Security Research Engineer at APISEC (San Francisco), brought the day’s most technical lens. Focused on APIs, the invisible connective tissue linking banks to fintech apps, hospitals to insurance platforms, and telecom providers to government databases.

Why APIs Matter More Now

APIs have quietly become the **largest and least-monitored attack surface** in most organizations, often larger than the traditional web application layer. A single exposed endpoint, an over-permissioned token, or an undocumented internal API can hand an attacker a direct path into systems that look secure from the outside.



Ms. Bandana Kaur

Security Research Engineer,
APISEC

Joining virtually from India

The Shared-Economy Risk

A breach does not need to originate inside an organization anymore. It can arrive through a vendor’s API, a partner integration, a payment gateway, or a third-party service that no one in the room thought to audit. In a financial or telecom context, an organization can do everything right internally and still be compromised through a connection point it did not build and does not monitor.

AI as an Amplifier

Most AI tools, particularly large language model and automation platforms pull and push data through API calls, often through the same infrastructure as everything else, but with less scrutiny. An AI feature bolted onto an existing product can quietly open new data pathways that were never threat-modeled.

Core argument: Securing a single organization is no longer sufficient when its APIs connect it to dozens of others, each with its own security maturity, blind spots, and AI integrations layered on top. Cross-sector defense must mean shared visibility into what is being exposed and consumed across those connections.

Cyber Animation: Ms. Maheen Ishfaq, Founder of CyberWithMaheen



Ms. Maheen Ishfaq

Founder, CyberWithMaheen

Ms. Maheen Ishfaq closed the pre-break segment with something entirely different: an animated short film making the same argument the keynotes had been making but through story instead of statistics.

The animation dramatized a scam call in progress: an OTP being requested, a system being hacked in real time, servers going down. Staged to feel like something that could happen to anyone watching.

The film then pulled back to the structural point underneath: Pakistan's critical sectors are fighting these threats alone. Banks do not talk to hospitals. Schools do not talk to startups. Governments sit cut off from all of it, five sectors, five walls, one storm.

Her reframe: what if, instead of five separate walls, the answer was **one shared roof**, a collective defense where nobody fights alone? That question led directly into the introduction of the Pakistan Cybersecurity Alliance itself.

"Whether someone is 15 or 65, they should know what's coming and how to stop it."

— **Ms. Maheen Ishfaq, Founder, CyberWithMaheen**

Speaker: Dr. Shah Nazir

“Shifting from SDLC to SSDLC”

Dr. Shah Nazir, Associate Professor at IIUI, brought the day’s most technical and structured session. Where earlier speakers made the case for *why* cybersecurity matters, Shah Nazir’s session answered a narrower, more practical question: **how does an organization actually build software that can survive an attack?**

Core distinction: *SDLC asks whether software works. Secure SDLC asks whether it works, and whether it can be trusted to hold out under attack.*



Dr. Shah Nazir

Associate Professor

International Islamic University, Islamabad

SSDLC Phase by Phase

Phase	Secure Practice
Requirements	Risk assessment and privacy/compliance checks done up front, not as an afterthought
Design	Formal threat modeling (STRIDE, DREAD); secure design patterns such as zero trust and least privilege
Coding	Static Analysis (SAST) and Software Composition Analysis (SCA), paired with mandatory secure-coding review
Testing	Dynamic Analysis (DAST) and Interactive Application Security Testing (IAST) alongside standard functional tests
Maintenance	Continuous monitoring, automated patch management, and a formal, usable incident-response plan

Table 5: The Secure Software Development Life Cycle, phase by phase.

What Skipping SSDLC Has Cost: Real-World Breaches

Incident	Year	Impact	Root Cause
Sony PlayStation Network	2011	77M users' data exposed; 24 days offline	Fundamental software security flaw
Mt. Gox	2014	850,000+ bitcoins lost; bankruptcy	Exploited security and software errors
Yahoo	2016	500M+ accounts exposed	Missing security update on portal
Equifax	2017	143M Americans' data; \$1.4B+ cost	Unpatched Apache Struts vulnerability
Microsoft Power Apps	2021	38 million records exposed	Insecure default settings
Discord.io	2023	Platform shut down entirely	Hacking event

Table 6: Breaches traced back to gaps that a Secure SDLC is designed to close.

International Speaker 2: Mr. Charles Jeremiah Nosa

“Tech-Sector Data Exfiltration Trails in Metadata”

**Mr. Charles Jeremiah Nosa**

Digital Forensics Investigator

Greyhawk Forensic and
Cybersecurity (Nigeria)

Mr. Charles Jeremiah Nosa, Digital Forensics Investigator at Greyhawk Forensic and Cybersecurity (Nigeria), brought the day's second international perspective, this time from the investigative side.

His session was built around a live digital forensics simulation demonstrating how **metadata: the data about data** can expose the trail of a data exfiltration event even when the file's content reveals nothing.

The evidence file used: a 1.4 GB disk image (Kimberley.E01) sourced from the AB-CW-DFIR repository on GitHub. Jeremiah adhered strictly to a documented Chain of Custody process throughout; the evidence file's integrity remained fully intact.

What the Metadata Revealed

The simulation produced findings across several categories:

- Web browser activity logs
- Email access patterns
- Document access history
- Timestamped user interactions
- Login frequency and authentication events
- Device-specific information

Key finding: An attacker can alter or delete the content of a stolen file, but metadata trails left behind in timestamps, access logs, and device signatures are considerably harder to erase cleanly. Metadata is the evidence the attacker cannot hide.

Glossary of Terms

Term	Definition
API	A set of rules that lets two software systems exchange data. Flagged as the main, often under-monitored, channel through which cross-sector cyber risk travels.
Attack surface	The total set of points through which a system could be attacked.
Chain of Custody	The documented, unbroken record of who handled a piece of evidence and when, used in digital forensics to prove the evidence was not altered.
DAST	Dynamic Application Security Testing: security testing performed on a running application.
IAST	Interactive Application Security Testing: analyzes an application from within while it runs.
LLM	Large Language Model: an AI model trained on large volumes of text; cited as a category of AI tool that opens new API data pathways with less security scrutiny.
National CERT	Pakistan's national body responsible for capacity building and coordinating responses to cybersecurity incidents.
NICAT	National Incubation Center for Aerospace Technology: Rawalpindi-based incubation center and co-host of the event.
NIST SSDF	NIST Secure Software Development Framework: organizes secure development practices into four groups.

Term	Definition
OWASP	Open Web Application Security Project: publishes the Top 10 list of critical web application risks and the SAMM maturity model.
PCA	Pakistan Cybersecurity Alliance: the free, industry-led community at the center of this report.
SAST	Static Application Security Testing: scans source code for vulnerabilities without running it.
SCA	Software Composition Analysis: identifies security risk of third-party and open-source components.
SDLC	Software Development Life Cycle: the standard five-phase process for building software.
Shift-left	A security methodology that moves testing and checks as early as possible in the development pipeline.
Social engineering	Manipulating people, rather than exploiting technical systems, to gain unauthorized access.
SSDLC	Secure Software Development Life Cycle: integrates security into every phase of the SDLC.
STRIDE	Threat-modeling framework (Spoofing, Tampering, Repudiation, Information disclosure, Denial of service, Elevation of privilege).
Supply-chain breach	A security breach originating through a vendor, partner, or supplier rather than directly within an organization.

Term	Definition
Wazuh	An open-source security monitoring platform referenced as widely adopted open-source security tooling in Pakistan.
WE Finance Code	A State Bank of Pakistan framework for women-led MSME financing, with MMBL among its 22 committed banks.
Zero trust	Security design principle assuming no user or system should be automatically trusted; access is limited to the minimum necessary.

Audience Voices — Questions & Insights from the Room

One of the defining features of Threat Horizons Pakistan is that the audience does not just listen, they contribute. The Q&A and interactive session produced some of the most candid, sector-specific observations of the day. The individuals below engaged directly with speakers and panelists, adding perspectives that no keynote alone could have surfaced.



“The most powerful moment in any event is when the audience stops being passive and starts contributing. That is when the room becomes a think-tank.”

— Pakistan Cybersecurity Alliance (PCA)

Closing Note — Ms. Fizza Malik, Vice President, Pakistan Cybersecurity Alliance

**Ms. Fizza Malik**

Vice President

Pakistan Cybersecurity
Alliance (PCA)Operationally managed
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“The umbrella starts with us.”

As we close Threat Horizons Pakistan Episode 1, I want to reflect on what this room actually built today because it was more than an event.

PCA was founded on a single belief: that Pakistan’s cybersecurity, governance, AI, and industry sectors have been operating in silos for too long. Today, for a few hours, that changed. Professionals from nine sectors sat in the same space, asked the same hard questions, and started building the same answers together.

The problems we identified today, fragmented incident response, an academic curriculum lagging behind industry need, open-source tools filling gaps that enterprise security should cover, and a digital economy growing faster than its defenses are real. They are well-documented. And they are solvable.

But they are only solvable if we keep showing up. Not just to events, but to the work between events. The partnerships, the referrals, the shared intelligence, the cross-sector conversations that continue after this hall empties.

Before we close, I am proud to share one milestone born from today’s session: **iSeeWaves and HerVoice have signed a Memorandum of Understanding, the first institutional partnership to emerge from PCA’s inaugural event.** This is what the umbrella looks like in practice.

Do not be the person who came, listened, and went home with nothing. Make the right connections. Collaborate. Build something worthy. Make impact. And come back next month because Episode 2 is already on its way.

“Threat Horizons Pakistan is not the end of a conversation. It is the beginning of an alliance. The umbrella starts with us. One nation. One defense. No sector stands alone.”

— Ms. Fizza Malik, Vice President, PCA

Acknowledgements

The Pakistan Cybersecurity Alliance (PCA) gratefully acknowledges the institutions whose support made Threat Horizons Pakistan possible.

- **iSeeWaves (Private) Limited:** For operating the event from conception to execution, and for anchoring the day with the opening keynote.
- **National Incubation Center for Aerospace Technologies - NICAT, Rawalpindi:** For hosting the event and providing the venue and facilities that brought together industry, academia, and policy under one roof.
- **All speakers and panelists:** For their time, candor, and contribution to Pakistan’s cybersecurity dialogue.

Episode 2 is Coming.

One room. Every sector's problem. Every sector's solution.

Pakistan Cybersecurity Alliance (PCA)

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